



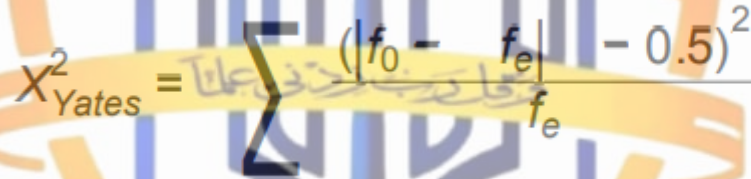
SMALL-SAMPLE TESTS OF INDEPENDENCE

Yates' Correction

- Yates' Correction is not a test as such but a means to correct a problem faced when using Pearson's Chi Squared test on 2×2 contingency tables.
- In small sizes where cell counts may be low (i.e., less than 10, less than 5) or relatedly in samples where there are cell counts below 10 or below 5, it is problematic as Pearson's Chi-Squared has a tendency to over-reject.
- The Yates' Correction adjusts for this problem.
- Like a Pearson's Chi-Squared test it allows us to identify an association between two independent, categorical variables, each of which is dichotomous.

Yates' Correction

In order to apply the Yates correction, **subtract .5** from the numerical difference between the observed frequencies and expected frequencies. The formula looks complicated, but it's just the Chi² formula with the 0.5 subtraction


$$\chi^2_{\text{Yates}} = \sum \frac{(|f_o - f_e| - 0.5)^2}{f_e}$$

CRITERIA FOR YATES CORRECTION

- Both variables are dichotomous qualitative (2 X 2 table).
- Sample size of ≥ 40 .
- One or more of the cells has expected value of < 5 .

Example

Table 2: Crosstabulation of Preferences for Soccer and Pies: Observed and Expected Counts.

	Frequencies	Likes Soccer	Dislikes Soccer	Totals
Likes Pies	Observed	9	5	14
	Expected	7.7	6.3	
Dislikes Pies	Observed	2	4	6
	Expected	3.3	2.7	
Totals		11	9	20

$$\begin{aligned}
 \chi^2_{Yates} &= \frac{(|9 - 7.7| - 0.5)^2}{7.7} + \frac{(|5 - 6.3| - 0.5)^2}{6.3} + \frac{(|2 - 3.3| - 0.5)^2}{3.3} + \frac{(|4 - 2.7| - 0.5)^2}{2.7} \\
 &= 0.083 + 0.102 + 0.194 + 0.626 \\
 &= 1.005
 \end{aligned}$$

Critical value = 3.84

Conclusion: Accept H_0

Fisher's Exact Test

Fisher's Exact Test is used to determine whether or not there is a significant association between two categorical variables. It is typically used as an alternative to the Chi-Square Test of Independence when one or more of the cell counts in a 2×2 table is less than 5.

Under H_0 : *independence*, conditioning on both sets of marginal totals yields the hypergeometric distribution

$$p(t) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t} \binom{n_{2+}}{n_{+1} - t}}{\binom{n}{n_{+1}}}.$$

This formula expresses the distribution of $\{n_{ij}\}$ in terms of only n_{11} .

Given the marginal totals, n_{11} determines the other three cell counts.

Fisher's exact test

Fisher's exact test consists first of generating all tables that are consistent with the given marginal totals

Then for each such table, we calculate the sums of the individual probabilities associated with tables that are as extreme as or more extreme than the observed table.

For 2X2 tables, independence is equivalent to the odd ratio = 1.

$$H_0: \theta = 1 \text{ vs. } H_a: \theta > 1$$

more extreme means "tables having larger n_{11} "

Thus P-value = $P(n_{11} \geq t_o)$, where t_o is the observed value of n_{11} .

FISHER'S EXACT TEST CRITERIA

- ❖ Sample size of < 20 .
- ❖ Sample size of 20 to < 40 but one of the cell has expected value of < 5 .

Formula for Fisher's Exact Test

$$\frac{(a+b)! (a+c)! (b+d)! (c+d)!}{N! a! b! c! d!}$$

Example

Distribution of Underweight and Normal Weight for Taxi Drivers and Bus Drivers

	Underweight	Normal	Total
Bus Drivers	8	11	19
Taxi Drivers	3	11	14
Total	11	22	33

There is an association between the prevalence of underweight and the type of vehicle driven by the public vehicle drivers.

In this analysis, it is a 2 X 2 table, cells with an expected value < 5 (4.67) and small sample size, therefore the best type of analysis possible is **Fisher's Exact Test**.

Calculations

$$\frac{(a+b)! (a+c)! (b+d)! (c+d)!}{N! a! b! c! d!}$$

$$\begin{aligned} p1 &= \frac{19!14!11!22!}{33!8!11!3!11!} \\ &= \frac{4.758 \times 10^{56}}{3.3471 \times 10^{57}} = 0.142 \end{aligned}$$

Create 3 more extreme tables by deducting 1 from the smallest value. Continue to do so till the cell becomes zero;

KB	N		KB	N		KB	N	
9	10	19	10	9	19	11	8	19
2	12	14	1	13	14	0	14	14
11	22	33	11	22	33	11	22	33

$p2 = 0.0434$
 $p3 = 0.00668$
 $p4 = 0.00039$

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- ▢ Total $p = 0.142 + 0.0434 + 0.00668 + 0.00039$
 $= 0.19247$
 - ▢ This is the p value for single-tailed test. To make it the p value for 2-tailed, times the value with 2; $p = 0.385$.
 - ▢ p is bigger than 0.05, therefore the null hypothesis is not rejected.
 - ▢ There is no association between occupation and UW ;-)

Yates' Correction Vs Fisher's Exact test

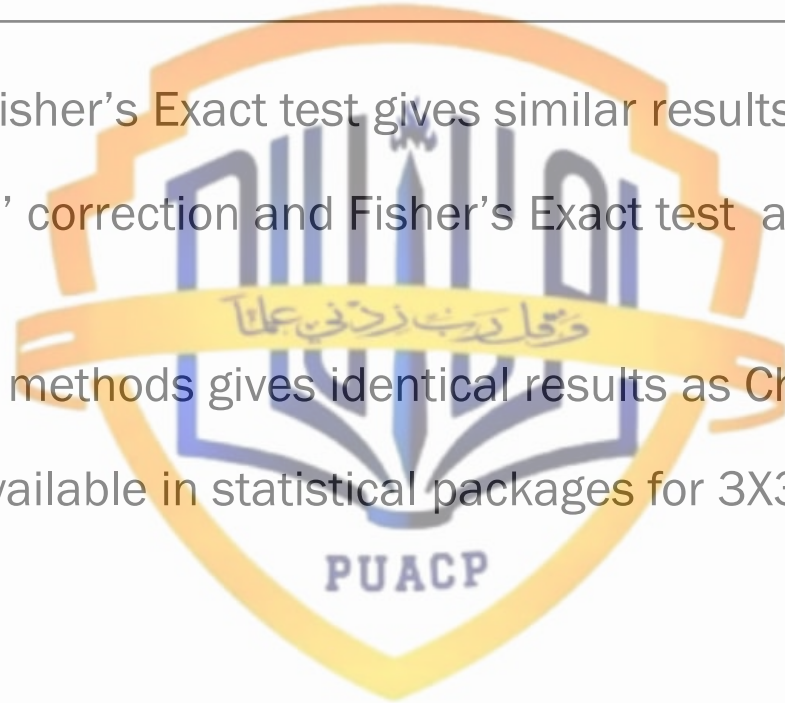
With small sample sizes, chi-square is not accurate, with or without Yates' correction. Fisher's exact test, as its name implies, always gives an exact P value and works fine with small sample sizes.

The Yates' continuity correction is designed to make the chi-square approximation better, but it overcorrects so gives a P value that is too large but Fisher's Exact test always gives an exact P value and works fine with small sample sizes.

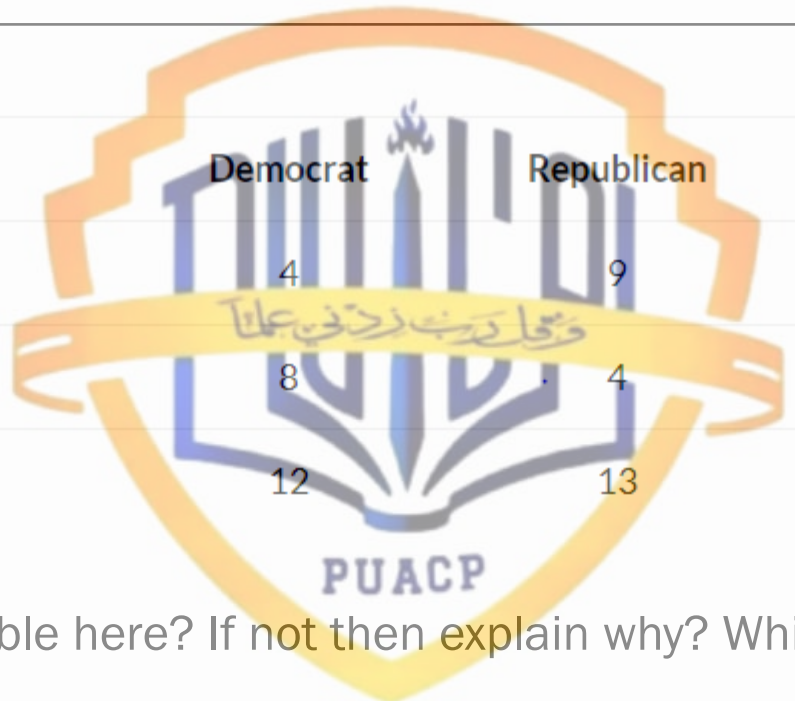
Fisher's test (unlike chi-square) is very hard to calculate by hand, but is easy to compute with a computer. Most statistical books advise using Fisher's exact test instead of chi-square test.

Important to Note

1. Yates' correction and Fisher's Exact test gives similar results.
2. P-value obtain by Yates' correction and Fisher's Exact test are higher than those calculated by Chi-Square test.
3. In large samples these methods gives identical results as Chi-square.
4. Fisher's exact test is available in statistical packages for 3X3, 4X4 contingency tables.



Practice Questions



The logo of the Punjab University Academic Council (PUACP) is centered in the background. It features a shield with a book and a flame, with the text 'PUACP' at the bottom and Urdu script in the middle.

	Democrat	Republican	Total
Male	4	9	13
Female	8	4	12
Total	12	13	25

1. Is chi square is applicable here? If not then explain why? Which test is appropriate here?
2. Test the hypothesis Gender and political party preference are independent.

Answer: p-value= 0.11523

	Not Inoculation	inoculation
Not Attacked	3	5
Attacked	10	2

Check the association between 2 variables.

	Psychotics	Neurotics
Suicidal	4	4
No Suicidal	16	16

Check the association between 2 variables